Moukthika Nellutla

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Education

University of Illinois Urbana-Champaign *Expected May 2028*
Bachelor of Computer Science and Bioengineering *GPA: 3.51/4.00*

- **Relevant Coursework**: Bioinformatics, Linear Algebra, Data Structures and Algorithms,Biochemistry, Database Systems, Computational Tools for Biological Data, Machine Learning, Computer Systems, Signals and Systems, Differential Equations
- **Certifications**: Break Through Tech Cornell University online certificate in Machine Learning Foundations *July 2026*

Technical Skills

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, R, MATLAB, HTML/CSS
Frameworks: React, FastAPI, Flask, Node.js
ML & Data Libraries: PyTorch, scikit-learn, TensorFlow, NumPy, Pandas, SciPy, OpenCV
Developer Tools & Cloud: Git, Docker, Jupyter, VS Code, Linux, AWS, GCP, Azure
Databases & Platforms: PostgreSQL, SQLite,Tableau, Stripe Checkout

Experience

Software Engineer October 2025 – May 2026
Gies College of Business *Champaign, IL*

- Architected an LLM inference platform used by **10+** faculty members for academic evaluation, building secure **REST APIs**, scalable backend services, **interactive analytics projection**, and product management dashboards.
- Optimized model inference pipelines for latency and computation efficiency, increasing business system reliability and throughput by **30%**.
- Streamlined **RAG** workflows with automated logging, supporting **100s** of LLM test runs.

Undergraduate Research Assistant August 2025 – Present
Sweedler Research Group *Champaign, IL*

- Quantifying **5k+ individual peptides** in rat dorsal root ganglia to study neuropathic pain using **LC–MS/MS data**.
- Authored Python/R scripts to **group overlapping** peptides by sequence position, sum signals, and differentiate unique vs. shared peptides.
- Planned **statistical comparisons via data science** across three animal groups (D, H, J) during acute pain stage to **identify biomarkers**.

Machine Learning Intern May 2025 – August 2025
Moffitt Cancer Center *Tampa, FL*

- Deployed NLP and **cross-functional machine learning techniques** to enhance clinical trial matching for **20k+ oncology patients** using **Deep6 AI** and Jupyter model workflows.
- Benchmarked **6+ ML models** (MLP, Logistic Regression, Gradient Boosting, KNN, SVM) using **confusion matrices** and performance metrics for data automation.
- Engineered scalable preprocessing and feature engineering pipelines that supported **100s** of patient–trial matches for downstream model training.

Community & Leadership

Director of NeuroHack Sept 2024 – Present
NeuroTech@UIUC *Champaign, IL*

- Led a neurotech-focused hackathon for **50+ participants**, securing partnerships with **ID8 and ElevenLabs** for sponsored prizes and managing end-to-end planning, logistics, and execution.
- Spearheaded NeuroBot, merging EEG preprocessing with **transformer-based NLP prompt** pipelines to generate personalized dashboards, **leading a 10–15 member team**.
- Implemented real-time EEG data acquisition modules with artifact rejection and visualization dashboards to support ongoing experiments.

Projects

MediPay | *Python, FastAPI, React, SQLAlchemy, SQLite* Feb 2026 – Mar 2026
GitHub: github.com/Mnellutla1120/hackillinois2026medicaldebt

- Built a full-stack medical debt repayment API at **HackIllinois 2026**, implementing a risk-scoring formula based on debt, income, and credit score to categorize risk into **3 levels**.
- Implemented **8 REST API endpoints** for debt CRUD operations, risk assessment, and repayment summaries, with interest-based amortization and configurable down payments; integrated Stripe Checkout for payments.

Demynsify | *Python, NLP, Machine Learning* Dec 2025 – Jan 2026
GitHub: github.com/Mnellutla1120/demynsify

- Developed a prototype for medical misinformation analysis using a dataset of **100,000+ articles and research papers** published between 1997 and 2022.
- Built a Python data pipeline to preprocess health-related text and prepare curated data for supervised machine learning experiments.